



# Modelling Financial Markets

## Chapter 10: Market Microstructure



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## Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AMM: Automated Market Maker
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CARA: Constant Absolute Risk Aversion
- ▣ CEX: Centralised EXchange
- ▣ CFTC: Commodity Futures Trading Commission
- ▣ CI: Confidence Interval
- ▣ DEX: Decentralised EXchange
- ▣ EDGE: Efficient Discrete Generalized Estimator (of the bid-ask spread)
- ▣ EKOP: Easley–Kiefer–O'Hara–Paperman (model of informed trading)
- ▣ ETF: Exchange-Traded Fund
- ▣ FOK: Fill-Or-Kill (order)
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GMM: Generalized Method of Moments
- ▣ GTC: Good-Till-Cancelled (order)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HFT: High-Frequency Trading
- ▣ IOC: Immediate-Or-Cancel (order)
- ▣ LLM: Large Language Model
- ▣ LOB: Limit Order Book
- ▣ MA: Moving Average
- ▣ MRR: Madhavan–Richardson–Roomans (structural model of trading costs)

## Acronyms used in this material (2/2)

- ▣ NYSE: New York Stock Exchange
- ▣ OHLC: Open, High, Low, Close
- ▣ OLS: Ordinary Least Squares
- ▣ PIN: Probability of INformed trading
- ▣ SEC: Securities and Exchange Commission
- ▣ TWAP: Time-Weighted Average Price
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VAR: Vector AutoRegression
- ▣ VECM: Vector Error-Correction Model
- ▣ VIX: CBOE Volatility Index
- ▣ VPIN: Volume-synchronised Probability of INformed trading
- ▣ VWAP: Volume-Weighted Average Price
- ▣ XRP: XRP (the native token of the XRP Ledger)

## Outline

Motivation

The Limit Order Book

The Bid-Ask Spread

Illiquidity and Price Impact

Intraday Patterns

Information and Prices

Price Impact and Optimal Execution

High-Frequency Trading and Market Stress

Crypto Market Microstructure

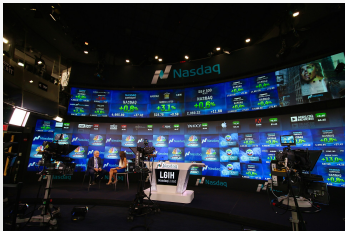
Liquidity at the Bucharest Stock Exchange

AI for Scientific Discovery

Summary

References

## From the Trading Floor to the Matching Engine



Nasdaq MarketSite, Times Square, New York: a studio,  
not a trading floor

Photo: Luca Marfè (2015); CC BY 2.0; Wikimedia Commons

- For two centuries prices were made by people in a room: brokers, specialists, open outcry
  - ▶ The specialist kept the order book on paper and quoted a bid and an ask
- Today almost all equity, futures and crypto trading runs on computers that match orders in microseconds
- The rules of the matching engine (priority, tick size, auctions) shape prices and costs [Gould et al. \(2013\)](#)
- This chapter: how a price is formed and what trading costs, and the econometrics that measures both
  - ▶ Textbooks: [Hasbrouck \(2007\)](#), [Bouchaud, Bonart, Donier & Gould \(2018\)](#), [Franke, Härdle & Hafner \(2019\)](#)

## Data Used in This Chapter

| Series                 | Content                                                                                  | Period    |
|------------------------|------------------------------------------------------------------------------------------|-----------|
| SPY, 5-minute bars     | ETF on the S&P 500; regular session 09:30–16:00 New York time; 1 477 full days           | 2020–2026 |
| Bitcoin, 5-minute bars | 24 hours a day, 7 days a week; 208 596 bars                                              | 2024–2026 |
| US stocks, daily       | SPY, Apple, Microsoft, Nvidia, JPMorgan, GameStop, Strategy                              | 2016–2026 |
| BVB stocks, daily      | Ten blue chips and the BET ETF; values in lei converted to USD at the BNR reference rate | 2016–2026 |
| Crypto-assets, daily   | Bitcoin, Ethereum, Solana, XRP, Dogecoin, Cardano, Litecoin                              | 2016–2026 |
| VIX, daily             | Implied volatility of the S&P 500                                                        | 2020–2026 |

- Daily data: open, high, low, close and volume; prices adjusted for dividends and splits; days without trades removed
- Order books in Section 2 and the bid-ask bounce in Section 3 are **simulated**, and labelled as such

## Liquidity and the Limit Order Book: Definitions

- **Liquidity**: trading a quantity quickly at a price close to fair value; no single number captures it
  - ▶ **Tightness** (the spread), **depth** (size needed to move the price), **resiliency** (speed of recovery) Kyle (1985); **immediacy** Harris (2003)
- **Limit order book** (LOB): all standing limit orders, by price and time of arrival Gould et al. (2013)
- Best bid  $b_t <$  best ask  $a_t$ ; mid-price  $m_t = (a_t + b_t)/2$ ; relative spread  $(a_t - b_t)/m_t$  in basis points (1 bp = 0.01%)
  - ▶ **Depth**: quantity waiting at a price level; **tick size**: smallest price step (USD 0.01 for most US stocks above USD 1)

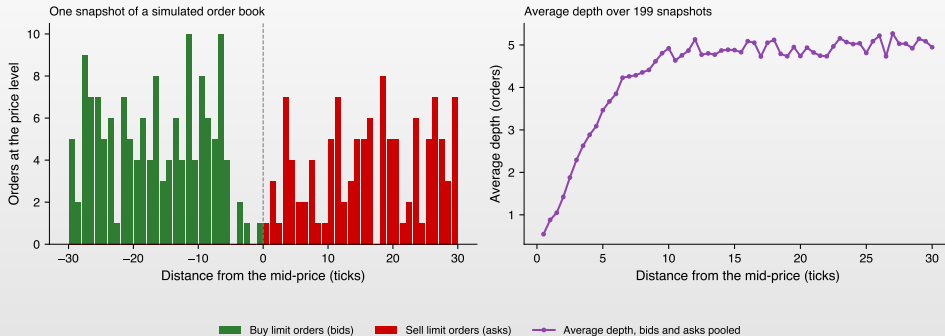
## Orders, Priority and Matching

- **Market order:** immediate execution against the available liquidity (within venue rules), uncertain price; **limit order:** price bounded by the limit, uncertain execution
  - ▶ Variants: IOC, FOK, GTC, stop and iceberg orders (glossary)
  - ▶ Resting (non-marketable) limit orders supply liquidity; market orders and marketable limit orders (a buy limit at or above the best ask) consume it
- **Price-time priority:** the better price first, then the earlier order at the same price (a queue); queue position is valuable (Section 8)
- **Call auctions** at the open and close: orders are pooled and matched at the single price that maximises the executed quantity
- A market buy of size  $Q$  walks up the ask levels: it pays the half-spread plus the cost of the deeper levels (the impact function below)

## A Simple Model of the Order Book

- **Zero-intelligence** model: orders arrive at random, with no strategy [Farmer, Patelli & Zovko \(2005\)](#)
  - ▶ Limit orders: rate  $\alpha$  per price level, placed uniformly up to 40 ticks from the opposite best quote
  - ▶ Market orders: rate  $\mu$  per side; cancellations: rate  $\delta$  per standing order
- Here  $\alpha = 0.1$ ,  $\mu = 0.5$ ,  $\delta = 0.02$ ; 300 000 events, unit orders
- Farmer, Patelli and Zovko show that such a model already predicts much of the cross-section of spreads on the London Stock Exchange
- Use: a laboratory where spread, depth and impact can be measured exactly

## A Simulated Order Book

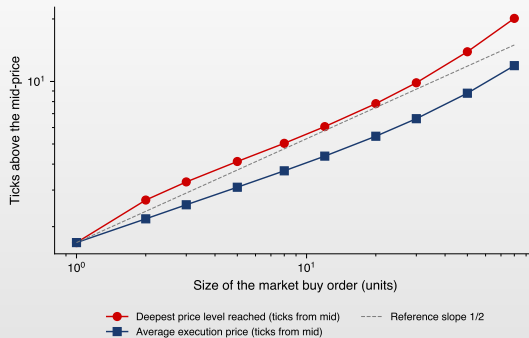


Left: one snapshot (bids green, asks red); right: average depth by distance from the mid, bids and asks pooled

## Reading the Simulated Order Book

- Mean spread 3.23 ticks, median 3; it exceeds 6 ticks 10% of the time
- Depth is thin near the mid and grows away from it
  - ▶ 0.8 orders at 1 tick, 4.8 at 10 ticks, 5.0 at 25 ticks (average over 199 snapshots)
- Reason: market orders eat the best levels first, so liquidity near the price is scarce [Tóth et al. \(2011\)](#)
- Real order books show the same rise away from the mid, often with a maximum a few ticks away [Bouchaud, Bonart, Donier & Gould \(2018\)](#)

## The Price Impact of a Market Order



Market buy orders of 1 to 80 units sent against each simulated snapshot; log-log axes

## Reading the Impact Function

- ▣ A 1-unit order reaches 1.7 ticks above the mid; an 80-unit order reaches 20.1 ticks
- ▣ **Concave** impact: slope 0.53 on log-log axes, close to  $1/2$ 
  - ▶ Average execution price: slope 0.44; 80 units cost 11.9 ticks on average
- ▣ Doubling the order does not double the cost: depth grows with distance from the mid
- ▣ Real markets show a similar concave law for large orders split over time (Section 7)

## Why Is There a Spread?

- A liquidity provider buys at the bid and sells at the ask; the spread pays three costs [Huang & Stoll \(1997\)](#)
- **Order processing:** fees, technology, clearing
- **Inventory:** holding an unwanted position exposes the provider to price risk
- **Adverse selection:** some counterparties know more [Bagehot \(1971\)](#)
  - ▶ The provider incurs losses when trading with informed traders and must recover them from uninformed traders (Section 6)

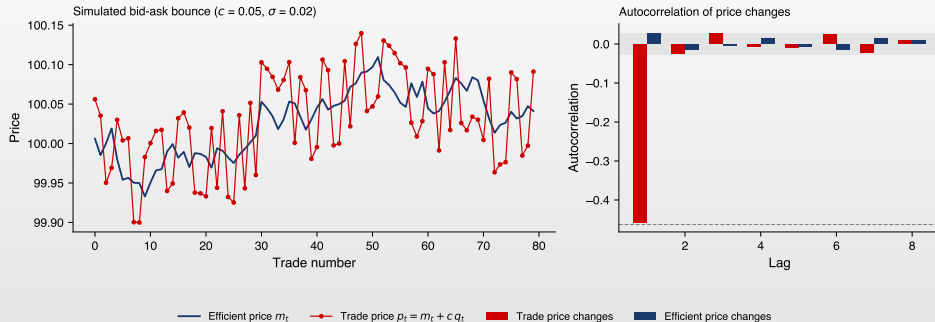
## Quoted, Effective and Realised Spread

- ▣ Trade at price  $p_t$ , direction  $q_t = +1$  (buyer-initiated) or  $-1$  (seller-initiated), mid-price  $m_t$
- ▣ **Quoted:**  $S_t^Q = (a_t - b_t)/m_t$ ; **effective:**  $S_t^E = 2q_t(p_t - m_t)/m_t$ , twice the signed distance from the mid (round-trip equivalent; the one-way cost is half)
- ▣ **Realised:**  $S_t^R = 2q_t(p_t - m_{t+\Delta})/m_t$ , the provider's revenue marked to the mid after  $\Delta$  (e.g. 5 minutes), before fees and inventory costs
  - ▶ Price impact  $= S_t^E - S_t^R = 2q_t(m_{t+\Delta} - m_t)/m_t$ : the lasting move of the mid after the trade, usually attributed to adverse selection, though inventory and order-flow pressure also contribute
- ▣ Quoted spread: quotes only; effective and realised spread: trades, the prevailing quotes and the trade direction, signed for example by the rule of [Lee & Ready \(1991\)](#)
- ▣ Without such data we **estimate** the spread from prices only: the next slides

## The Roll Model

- Efficient price: random walk  $m_t = m_{t-1} + \varepsilon_t$ ,  $\varepsilon_t \sim (0, \sigma^2)$  i.i.d. [Roll \(1984\)](#)
  - ▶ Trade price  $p_t = m_t + c q_t$ , with half-spread  $c$  and  $q_t = \pm 1$  with probability  $1/2$ , i.i.d. over time and independent of  $\varepsilon$
- Price change:  $\Delta p_t = \varepsilon_t + c(q_t - q_{t-1})$
- $\text{Var}(\Delta p_t) = \sigma^2 + 2c^2$ ,  $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$
- **Roll estimator:**  $\hat{s} = 2\hat{c} = 2\sqrt{-\widehat{\text{Cov}}(\Delta p_t, \Delta p_{t-1})}$ 
  - ▶ The bid-ask bounce creates negative autocorrelation even when the efficient price is a random walk

## The Bid-Ask Bounce (Simulated)



- 5 000 simulated trades with  $c = 0.05$  and  $\sigma = 0.02$ ; grey band:  $\pm 1.96/\sqrt{n}$ ; dashed line: the theoretical first-order autocorrelation

## Reading the Roll Simulation

- ▣ True spread  $2c = 0.10$ ; Roll estimate from the simulated trades:  $\hat{s} = 0.0998$
- ▣ First-order autocorrelation  $-0.458$ , theory  $-c^2/(\sigma^2 + 2c^2) = -0.463$
- ▣ No autocorrelation at lags 2 and beyond: the bounce lasts only one trade
- ▣ The efficient price changes are uncorrelated: the market is efficient, yet trade prices are not a random walk

## When the Roll Estimator Fails

- With real data the sample covariance is often **positive**:  $\hat{s}$  does not exist
  - ▶ SPY, 5-minute bars: the covariance within the day is positive on 41% of days
- Causes: momentum or news trends, orders split into many trades, stale prices, bars instead of trades
- With daily data the bounce (a few bp) is buried under the daily volatility (103 bp for SPY over the last two years)
- Hence estimators based on the daily high and low: the next two slides

## Statistical Properties of the Roll Estimator

- ▣  $\hat{\gamma}_1$ : sample autocovariance of  $\Delta p$  from  $T$  price changes (demeaned);  $\gamma_0 = \sigma^2 + 2c^2$ ,  $\gamma_1 = -c^2$  Harris (1990)
  - Bias from demeaning:  $\mathbb{E}[\hat{\gamma}_1] \approx \gamma_1 - (\gamma_0 + 2\gamma_1)/T$ , negative even when  $c = 0$
- ▣ Gaussian approximation (Bartlett's formula for an MA(1)):  $T \text{Var}(\hat{\gamma}_1) \rightarrow \gamma_0^2 + 3\gamma_1^2$
- ▣ Hence, ignoring the demeaning bias,  $P(\hat{\gamma}_1 > 0) \approx \Phi(\sqrt{T} \gamma_1 / \sqrt{\gamma_0^2 + 3\gamma_1^2}) \approx \Phi(\sqrt{T} \gamma_1 / \gamma_0)$  when  $c^2 \ll \sigma^2$ : close to 1/2 whenever  $c^2 \ll \sigma^2 / \sqrt{T}$
- ▣ Delta method:  $\text{se}(\hat{s}) \approx \sqrt{(\gamma_0^2 + 3\gamma_1^2)/T} / c$ ;  $\hat{s} = 2\sqrt{-\hat{\gamma}_1}$  is also biased (Jensen's inequality, truncation at 0)
- ▣ Seminar A1–A2 derive these moments and the standard error step by step

## SPY: Do Positive Covariances Reject the Roll Model?

- ▣ Per day, 77 five-minute returns; a one-tick spread means a half-spread  $c = 0.10$  bp, against a bar standard deviation of 9.2 bp ( $c/\sigma = 0.011$ )
- ▣ Monte Carlo under the Roll model with a one-tick spread (200 samples of the same size):  $\hat{\gamma}_1 > 0$  on 45.3% of days, 95% band [43.0, 48.1]%
  - ▶ Observed: 40.8%, below the band: the data are **more** negatively autocorrelated than a one-tick bounce, not less
- ▣ With no spread at all ( $c = 0$ ) the pooled Roll estimate is still 2.06 bp [1.50, 2.53]: most of the observed 2.82 bp is demeaning bias
- ▣ Bias-corrected: 1.92 bp, still about ten ticks: the remaining negative autocorrelation must come from sources other than the bounce, such as transient price pressure

## A Structural Model of Trading Costs: MRR

- Roll assumes i.i.d. trade signs; with first-order Markov signs,  $\text{Corr}(q_t, q_{t-k}) = \rho^k$ , independent of  $\varepsilon$  and without information,  $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2(1 - \rho)^2$ : Roll underestimates  $c$
- MRR model [Madhavan, Richardson & Roomans \(1997\)](#):  $p_t = m_t + \phi q_t + \xi_t$ ,  
 $m_t = m_{t-1} + \theta (q_t - \mathbb{E}[q_t | q_{t-1}]) + \varepsilon_t$ 
  - ▶  $\theta$ : permanent (adverse-selection) impact of the order-flow **surprise**;  $\phi$ : transitory cost (processing, inventory)
  - ▶  $q_t$  a first-order Markov chain; public news  $\varepsilon_t$  and rounding noise  $\xi_t$  i.i.d., mean zero, independent of each other and of  $q$
- With  $u_t = \Delta p_t - (\phi + \theta)q_t + (\phi + \rho\theta)q_{t-1}$ : GMM on  $\mathbb{E}[q_t q_{t-1} - \rho q_{t-1}^2] = 0$ ,  $\mathbb{E}[u_t - a] = 0$ ,  $\mathbb{E}[(u_t - a)q_t] = 0$ ,  $\mathbb{E}[(u_t - a)q_{t-1}] = 0$ ,  $a$  the drift (constant) of  $u_t$
- Implied effective spread  $2(\phi + \theta)$ ; adverse-selection share  $\theta/(\phi + \theta)$
- Needs signed trades (trade-and-quote data); our public 5-minute bars do not identify  $q_t$ , so MRR is not estimated here

## The Corwin–Schultz Estimator

- Idea: the daily high is usually a buy at the ask, the low a sell at the bid; the high-low range = volatility + spread [Corwin & Schultz \(2012\)](#)
- The volatility part grows with the length of the interval, the spread part does not: compare one day with two days
- $\beta = \sum_{j=0}^1 \left[ \ln \frac{H_{t+j}}{L_{t+j}} \right]^2$ ,  $\gamma = \left[ \ln \frac{H_{t,t+1}}{L_{t,t+1}} \right]^2$ ,  $\alpha = \frac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma}{3 - 2\sqrt{2}}}$
- $\hat{S} = \frac{2(e^\alpha - 1)}{1 + e^\alpha}$ , with  $H_{t,t+1}$ ,  $L_{t,t+1}$  the high and low over the two days
  - ▶ Negative two-day estimates are set to 0, then averaged over the period
  - ▶ The derivation assumes no price move overnight; for daily data the paper shifts  $H_{t+1}$ ,  $L_{t+1}$  down by  $L_{t+1} - C_t$  if  $L_{t+1} > C_t$ , up by  $C_t - H_{t+1}$  if  $H_{t+1} < C_t$  ( $C_t$  the close)

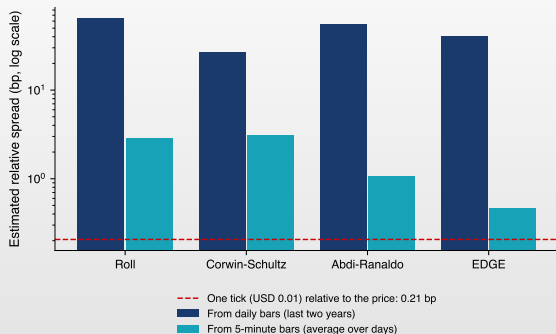
## The Abdi–Ranaldo Estimator

- ▣ Uses the close  $c_t$  and the mid-range  $\eta_t = (\ln H_t + \ln L_t)/2$  as a proxy for the mid-price [Abdi & Ranaldo \(2017\)](#)
- ▣  $\hat{S}^2 = 4 \mathbb{E}[(c_t - \eta_t)(c_t - \eta_{t+1})]$ , with  $c_t$  the log close
- ▣ The close sits at the bid or the ask; the distance to the neighbouring mid-ranges reveals the half-spread
- ▣ If the sample mean is negative, the estimate is set to 0
- ▣ For crypto-assets, both high-low estimators track the time variation of high-frequency liquidity well [Brauneis, Mestel, Riordan & Theissen \(2021\)](#)
  - ▶ The same study finds the Amihud ratio better for **levels** and for comparing venues

## The EDGE Estimator

- ▣ EDGE (Efficient Discrete Generalized Estimator) uses all four prices: open, high, low and close [Ardia, Guidotti & Kroencke \(2024\)](#)
- ▣ Two moment conditions identify  $S^2$  from products of log-returns between the open, the close and the mid-range  $\eta_t$ 
  - ▶ Each gives an unbiased estimate  $\hat{e}_k$  with variance  $v_k$ ; EDGE combines them optimally:  
$$\hat{S}^2 = (v_2 \hat{e}_1 + v_1 \hat{e}_2) / (v_1 + v_2)$$
- ▣ Minimum variance among estimators built from these moments, and unbiased under discrete observation of the efficient price
- ▣ Averaging over windows: the authors recommend signed estimates with negative values reset to zero, since more negative estimates tend to go with larger spreads [Ardia, Guidotti & Kroencke \(2024\)](#)

## Spread Estimators at Two Frequencies: SPY

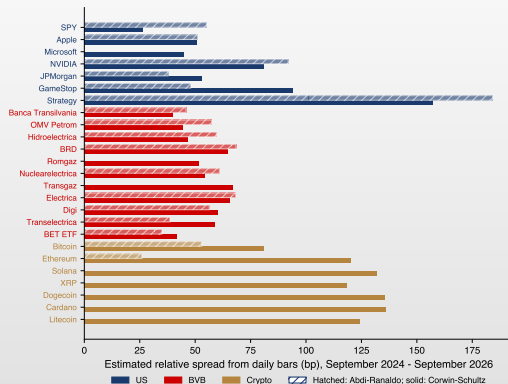


- Daily bars: September 2024 – September 2026; 5-minute bars, 1 477 days (2020–2026): Roll and Abdi–Rinaldo from the daily moments averaged over days, then the square root; Corwin–Schultz and EDGE (negatives set to 0) averaged over days

## Reading the Frequency Chart

- The **quoted** spread cannot be smaller than one price step (USD 0.01, i.e. 0.21 bp for SPY); an effective spread can, through price improvement
- From daily bars: Roll 64.3 bp, Corwin–Schultz 26.5 bp, Abdi–Rinaldo 55.1 bp
  - ▶ Two orders of magnitude above one tick for one of the most traded securities: implausible as a cost, consistent with volatility dominating the daily range [Jahan-Parvar & Zikes \(2023\)](#)
- From 5-minute bars: Roll 2.82, Corwin–Schultz 3.11, Abdi–Rinaldo 1.06 bp: closer, still above the tick
  - ▶ EDGE: 40.1 bp from daily bars, but 0.46 bp from 5-minute bars (95% CI resampling days [0.42, 0.49]): the closest to one tick
  - ▶ One tick is a plausibility benchmark, not a validation: that needs effective spreads from matched trades and quotes

## Spread Estimates Across 25 Assets



□ Daily bars, September 2024 – September 2026; BVB stocks in lei, prices adjusted for dividends and splits

## Reading the Cross-Section of Spread Estimates

- Median Corwin–Schultz estimate: US 53 bp, BVB 54 bp, crypto 124 bp
  - ▶ BVB blue chips look as liquid as US large caps: implausible, since they trade a thousand times less value (next section)
- The ranking follows volatility: crypto-assets and the most volatile stocks get the widest estimates
- Abdi–Rinaldo is 0 for 8 of the 25 assets: a negative sample mean
- Levels need validation against effective spreads [Jahan-Parvar & Zikes \(2023\)](#); for crypto-assets, [Brauneis, Mestel, Riordan & Theissen \(2021\)](#) find these estimators reliable for changes over time rather than for levels

## Recap: Order Book and Spreads

- Even random order flow creates a spread, a depth profile that rises away from the mid and concave impact [Farmer, Patelli & Zovko \(2005\)](#)
- The spread pays for order processing, inventory risk and adverse selection
- Bid-ask bounce makes trade prices negatively autocorrelated: the Roll estimator
- In small samples the Roll covariance is biased and often positive even when the model is true [Harris \(1990\)](#)
- High-low estimators work at high frequency; on daily data they are dominated by volatility when the spread is small relative to it [Jahan-Parvar & Zikes \(2023\)](#); EDGE is the efficient OHLC estimator

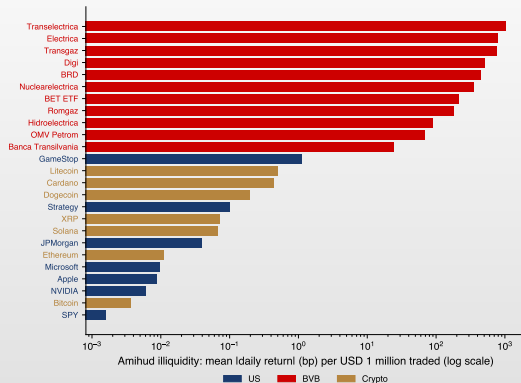
## The Amihud Illiquidity Ratio

- Daily price impact per unit of money traded [Amihud \(2002\)](#)

► 
$$\text{ILLIQ}_i = \frac{1}{D_i} \sum_{d=1}^{D_i} \frac{|r_{i,d}|}{\text{DVOL}_{i,d}}, \text{ with } r_{i,d} \text{ the daily return and } \text{DVOL}_{i,d} \text{ the traded value}$$

- Units here: basis points of price move per USD 1 million traded
- Needs only daily returns and volumes: available for every market and every period
- A low-frequency proxy of the Kyle price-impact coefficient (Section 6) [Hasbrouck \(2009\)](#)
- Traded value of BVB stocks: value in lei divided by the BNR reference rate of the day; crypto: traded value in USD summed across exchanges

## Amihud Illiquidity Across 25 Assets



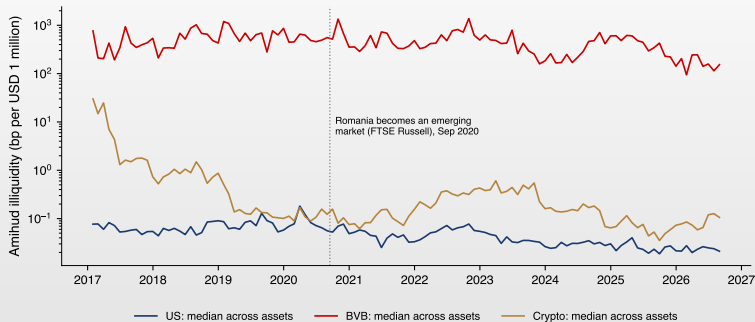
September 2024 – September 2026; log scale; each bar is the mean over about 500 trading days (730 for crypto)



## Reading the Amihud Ratios

- ▣ SPY: 0.0016 bp per USD 1 million; median daily value USD 40 124 million
  - ▶ Bitcoin: 0.0037 bp, with USD 41 834 million a day across exchanges
- ▣ Banca Transilvania, the most liquid BVB stock: 24.7 bp; median value USD 3.90 million a day
  - ▶ About 15 541 times the SPY figure; Transelectrica: 1024.0 bp
- ▣ Three clusters: US large caps and the top crypto-assets, then smaller crypto-assets and GameStop, then the BVB
- ▣ Unlike the spread estimators, the Amihud ratio separates markets by orders of magnitude

## Illiquidity over Time: Three Markets

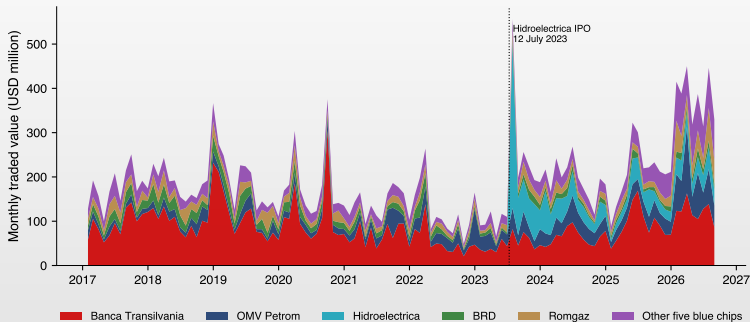


□ Monthly Amihud ratio, median across the assets of each group (without SPY and the BET ETF); log scale

## Reading Illiquidity over Time

- ▣ Crypto: from 7.6 bp per USD 1 million in 2017 to 0.089 in 2026
  - ▶ Crypto trading grew from a niche to institutional volumes; spot ETFs followed (Chapter 16)
- ▣ BVB: from 435 to 169: more liquid, but still about 7 212 times the US median
- ▣ US: from 0.062 to 0.023: large caps became even cheaper to trade
- ▣ Romania moved from frontier to emerging-market status in September 2020 [BVB \(history\)](#); the BVB median shows no break at that date

## Traded Value of BVB Blue Chips



- Monthly sum of daily traded value, ten blue chips, converted to USD at the BNR reference rate

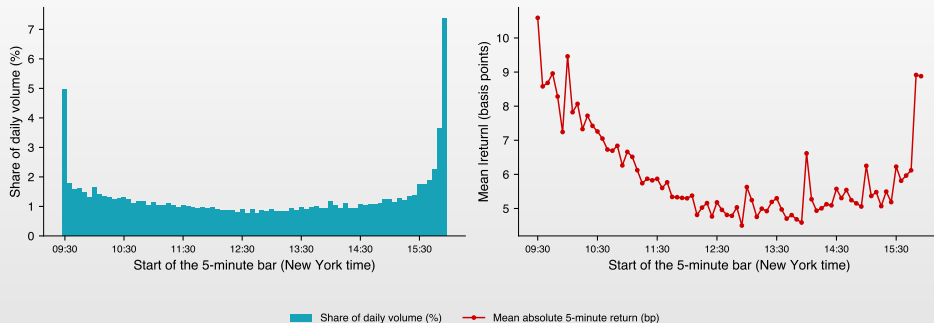
## Reading the BVB Traded Value

- ▣ Ten blue chips: USD 2 098 million traded in 2017, 1 494 million in 2022, 2 520 million in 2025
- ▣ Concentration: Banca Transilvania alone made 44% of the 2025 value, Hidroelectrica 13%
  - ▶ The Hidroelectrica listing of July 2023 added a second large, liquid stock
- ▣ One day of SPY trading exceeds a whole year of the ten BVB blue chips
- ▣ Consequence for a fund: a position of a few million USD takes days to build or unwind

## Is Liquidity Priced?

- **Level:** investors demand a higher return for assets that are costly to trade [Amihud & Mendelson \(1986\)](#)
  - ▶ Expected return increases with the spread, and concavely: long-horizon investors hold the illiquid assets
- **Time series:** expected market illiquidity raises future returns; an unexpected rise lowers current prices [Amihud \(2002\)](#)
- **Liquidity risk:** stocks that fall when market liquidity dries up earn a premium [Pástor & Stambaugh \(2003\)](#), [Acharya & Pedersen \(2005\)](#)
- Seminar Part C tests the time-series prediction on the BVB, US and crypto markets

## SPY Within the Trading Day



- 78 five-minute bars per day, 1 477 full days (October 2020 – September 2026); the first bar's return starts at the opening price

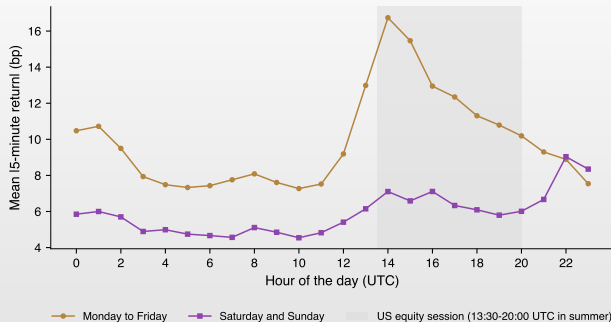
## Reading the U-Shape

- ▣ Volume: the first bar carries 5.0% of the day, a midday bar 0.9%; the last 30 minutes 18.6% (95% CI resampling days [18.3, 19.0]) against 7.7% if even
- ▣ Mean  $|r|$ : 10.6 bp in the first bar against 5.1 bp at midday; ratio 2.06, 95% CI [1.98, 2.15]: a systematic pattern, not a few extreme days
- ▣ Resample whole days: bars of one day share its volatility (seminar B1); spikes at 10:00 and 14:00 are scheduled releases

## Why a U-Shape?

- Documented for NYSE stocks by [Wood, McInish & Ord \(1985\)](#): returns and volume are highest at the open and the close
- **Open**: overnight news is absorbed; uncertainty is highest
  - ▶ Informed traders act early, before the information spreads
- **Clustering**: uninformed traders who can choose the time to trade together, and liquidity attracts liquidity [Admati & Pfleiderer \(1988\)](#)
- **Close**: index funds and ETFs trade at the closing price, which is fixed by the closing auction
- Practical rule: an execution algorithm follows the volume curve to reduce impact (Section 7)

## Bitcoin Around the Clock

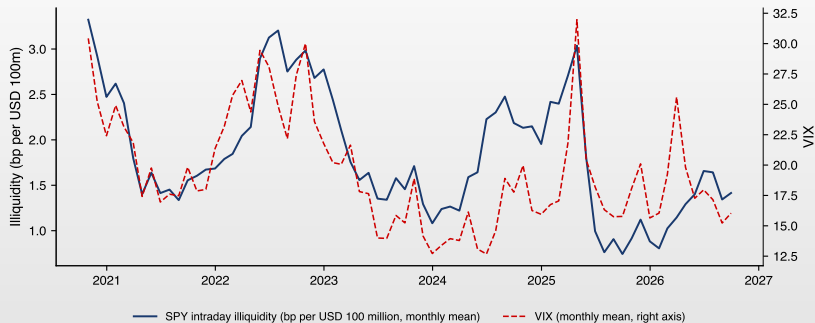


□ Mean absolute 5-minute return by hour (UTC), September 2024 – September 2026; 208 596 bars

## Reading the Bitcoin Clock

- Weekdays: peak of 16.7 bp at 14:00 UTC (US open and macro releases), low of 7.3 bp at 10:00 UTC
- Weekend: 5.9 bp against 9.9 bp on weekdays, 1.69 times calmer; seminar B7 attaches day-bootstrap intervals to both contrasts
- A 24/7 market still follows the clock of banks, ETFs and traditional investors; high-frequency Bitcoin data also help forecast daily risk [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)

## Liquidity and Volatility: SPY and the VIX



- Daily intraday illiquidity: mean of  $|r|$  / traded value over the 78 five-minute bars, in bp per USD 100 million; monthly means

## Reading Liquidity and Volatility

- ▣ Correlation of daily illiquidity with the VIX: 0.58; of monthly means: 0.69
- ▣ Regression of log illiquidity on log VIX: elasticity 0.95 (HAC standard error 0.12)
- ▣ April 2025: illiquidity 3.04 against a sample mean of 1.83
  - ▶ Liquidity providers widen quotes and shrink depth exactly when investors most need to trade
- ▣ Liquidity risk is therefore correlated with market risk [Acharya & Pedersen \(2005\)](#)

## Recap: Illiquidity and Intraday Patterns

- The Amihud ratio separates markets by orders of magnitude: BVB blue chips need thousands of times more price move per dollar than US large caps
- Volume and volatility are U-shaped over the trading day, with spikes at scheduled news
- Bitcoin, open 24/7, is most active when US markets are open and calmest at weekends
- Illiquidity rises with volatility: liquidity disappears when it is most needed

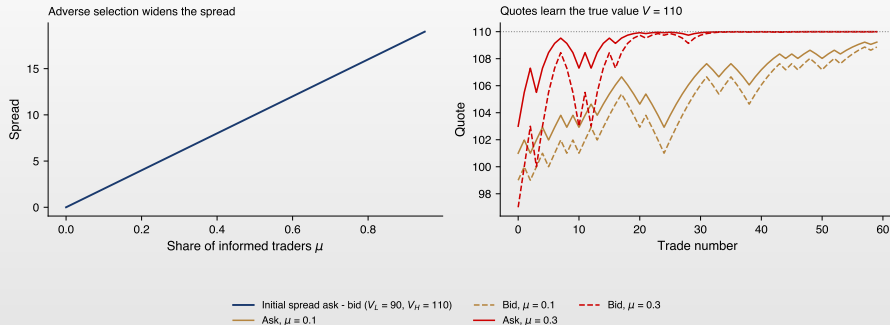
## The Glosten–Milgrom Model

- Value  $V \in \{V_L, V_H\}$ ,  $P(V = V_H) = \theta$ ; each period one trader arrives [Glosten & Milgrom \(1985\)](#)
  - ▶ With probability  $\mu$  informed (buys if  $V = V_H$ , sells if  $V = V_L$ ), otherwise uninformed, trading for liquidity needs (buys or sells with probability  $1/2$ ) [Bagehot \(1971\)](#)
- Competitive, risk-neutral market maker: zero expected profit on each trade
- $a = \mathbb{E}[V \mid \text{buy}]$ ,  $b = \mathbb{E}[V \mid \text{sell}]$
- $P(\text{buy} \mid V_H) = \mu + \frac{1-\mu}{2}$ ,  $P(\text{buy} \mid V_L) = \frac{1-\mu}{2}$
- After each trade, Bayes' rule updates  $\theta$ : a buy raises it, a sell lowers it

## Glosten–Milgrom: the Spread

- $P(V_H \mid \text{buy}) = \frac{\left(\mu + \frac{1-\mu}{2}\right) \theta}{\left(\mu + \frac{1-\mu}{2}\right) \theta + \frac{1-\mu}{2}(1-\theta)}, \quad a = V_L + (V_H - V_L) P(V_H \mid \text{buy})$
- At  $\theta = 1/2$ :  $a - b = \mu(V_H - V_L)$ 
  - ▶ The spread grows linearly with the share of informed traders
  - ▶  $V_L = 90$ ,  $V_H = 110$ : spread 2 for  $\mu = 0.1$ , 6 for  $\mu = 0.3$
- No inventory, no fees: the spread exists purely because of information
- Trade prices form a martingale: the market is efficient in the semi-strong sense (Chapter 2)

## Glosten–Milgrom in Action



- Left: initial spread as a function of  $\mu$ ; right: one simulated sequence of 60 trades with true value  $V = 110$  (same random draws for both  $\mu$ )

## Reading the Learning Paths

- With  $\mu = 0.3$  the bid exceeds 108 after 7 trades; with  $\mu = 0.1$  only after 48
- More informed traders: wider spreads, but faster price discovery
  - ▶ The trade-off between liquidity and informational efficiency
- The spread narrows as the market maker learns: less remains to be learned from the next trade
- Empirical counterpart: the probability of informed trading (PIN) [Easley, Kiefer, O'Hara & Paperman \(1996\)](#)

## The Kyle Model

- One insider knows  $v \sim N(p_0, \sigma_v^2)$  and submits a market order  $x$  [Kyle \(1985\)](#)
  - ▶ Noise traders submit  $u \sim N(0, \sigma_u^2)$ , independent of  $v$
  - ▶ The market maker sees only the total order flow  $y = x + u$  and sets  $p = \mathbb{E}[v \mid y]$
- The insider chooses  $x$  to maximise  $\mathbb{E}[(v - p)x]$ , knowing that his own order moves the price
- Linear equilibrium:  $x = \beta(v - p_0)$ ,  $p = p_0 + \lambda y$

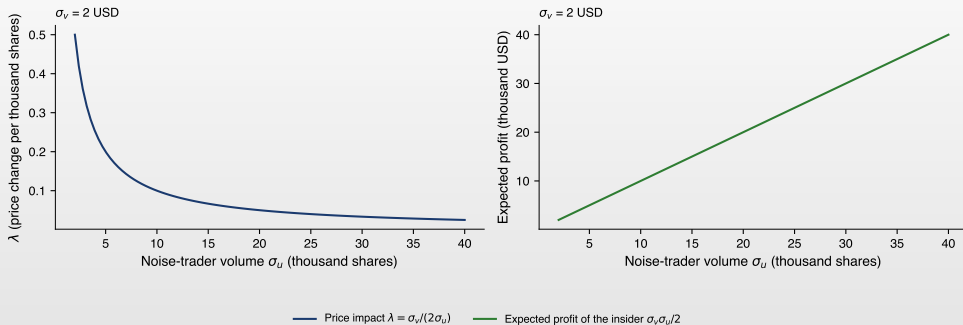
## Deriving the Kyle Equilibrium

- **Insider:** given  $p = p_0 + \lambda(x + u)$ , maximise  $\mathbb{E}[(v - p_0 - \lambda x - \lambda u)x \mid v] = (v - p_0)x - \lambda x^2$ 
  - ▶ First-order condition:  $x = (v - p_0)/(2\lambda)$ , so  $\beta = 1/(2\lambda)$ : he trades less aggressively because his own order moves the price
- **Market maker:** given  $x = \beta(v - p_0)$ ,  $(v, y)$  is jointly Normal, so  $\mathbb{E}[v \mid y]$  is the linear projection
  - ▶  $\lambda = \frac{\text{Cov}(v, y)}{\text{Var}(y)} = \frac{\beta\sigma_v^2}{\beta^2\sigma_v^2 + \sigma_u^2}$  (the population OLS slope of  $v$  on  $y$ )
- Substitute  $\beta = 1/(2\lambda)$ :  $\beta^2\sigma_v^2 = \sigma_u^2$ , hence  $\beta = \sigma_u/\sigma_v$  and  $\lambda = \sigma_v/(2\sigma_u)$
- Posterior variance  $\text{Var}(v \mid y) = \sigma_v^2 - \lambda^2\text{Var}(y) = \sigma_v^2 - \frac{\sigma_v^2}{4\sigma_u^2} 2\sigma_u^2 = \sigma_v^2/2$

## Kyle: the Equilibrium

- $\beta = \frac{\sigma_u}{\sigma_v}, \quad \lambda = \frac{\sigma_v}{2\sigma_u}, \quad \mathbb{E}[\text{profit of the insider}] = \frac{\sigma_v \sigma_u}{2}$
- $\lambda$  ("Kyle's lambda"): price move per unit of order flow;  $1/\lambda$  is **market depth**
  - ▶ More noise trading ( $\sigma_u$  up): deeper market, larger insider trades, larger insider profit
- $\text{Var}(v \mid p) = \sigma_v^2/2$ : the price reveals exactly half of the private information
- The insider hides behind noise traders: his order has the same variance as their orders,  $\text{Var}(x) = \sigma_u^2$

## Kyle's Lambda and the Insider's Profit



□  $\sigma_v = 2$  USD;  $\sigma_u$  from 2 000 to 40 000 shares

## From Models to Measurement

- Kyle's  $\lambda$  is estimated by regressing price changes on signed order flow; the Amihud ratio is its daily proxy [Hasbrouck \(2009\)](#)
- PIN: estimated from daily counts of buys and sells, by maximum likelihood [Easley, Kiefer, O'Hara & Paperman \(1996\)](#)
- VPIN: order imbalance in volume time, proposed as a measure of flow "toxicity" [Easley, López de Prado & O'Hara \(2012\)](#)
  - Contested: correctly computed, VPIN did not predict the Flash Crash and is largely a transformation of volume and volatility [Andersen & Bondarenko \(2014\)](#)
- Order-flow imbalance at the best quotes explains most short-horizon price changes [Cont, Kukanov & Stoikov \(2014\)](#)

## Estimating PIN, and Why It Is Fragile

- EKOP model [Easley, Kiefer, O'Hara & Paperman \(1996\)](#): each day an information event occurs with probability  $\alpha$  (bad news with probability  $\delta$ ); given the day's information state, daily buy and sell counts  $B$  and  $S$  are independent Poisson
  - ▶ Rates: uninformed  $\varepsilon_b, \varepsilon_s$ ; informed  $\mu$  added to the buy side on good-news days, to the sell side on bad-news days
- $L = (1 - \alpha) \pi(B; \varepsilon_b) \pi(S; \varepsilon_s) + \alpha \delta \pi(B; \varepsilon_b) \pi(S; \mu + \varepsilon_s) + \alpha(1 - \delta) \pi(B; \mu + \varepsilon_b) \pi(S; \varepsilon_s)$ ,  $\pi$  the Poisson pmf;
 
$$\text{PIN} = \frac{\alpha \mu}{\alpha \mu + \varepsilon_b + \varepsilon_s}$$
- Numerics: for liquid stocks ( $B \approx 10^4$ )  $e^{-\varepsilon} \varepsilon^B / B!$  under- or overflows; factor out common terms and use log-sum-exp [Lin & Ke \(2011\)](#)
  - ▶ Without it, estimates pile up at boundary values; the likelihood is also multimodal: use many starting values (seminar A9–A10)

## The Hasbrouck Trade–Quote VAR

- VAR (vector autoregression) in the mid-quote return  $r_t$  and the signed trade  $x_t$  [Hasbrouck \(1991\)](#)
  - ▶  $r_t = \sum_{i \geq 1} a_i r_{t-i} + \sum_{i \geq 0} b_i x_{t-i} + \varepsilon_{1t}, \quad x_t = \sum_{i \geq 1} c_i r_{t-i} + \sum_{i \geq 1} d_i x_{t-i} + \varepsilon_{2t}$
- Identification: the trade comes first within the interval ( $b_0$  enters, the reverse contemporaneous term does not): a recursive (Cholesky) ordering
- **Permanent impact:** the long-run cumulative response of the quote to a trade shock,  $\lim_{h \rightarrow \infty} \sum_{k=0}^h \partial r_{t+k} / \partial \varepsilon_{2t}$ : the information content of trades
- Variance decomposition of the efficient price: the share explained by trades generalises the adverse-selection share of MRR
- Like MRR, it needs signed trades; in Kyle's model the permanent impact is exactly  $\lambda$

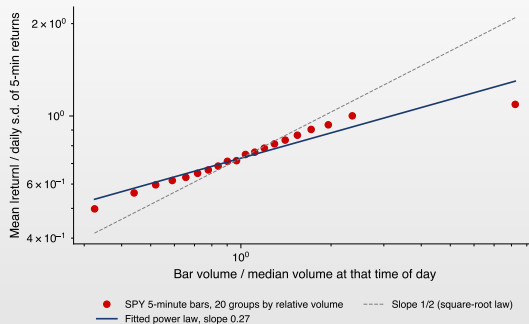
## Order Flow Is Long-Memory and Self-Exciting

- Signs of market orders are autocorrelated with a power-law decay  $\text{Corr}(\varepsilon_t, \varepsilon_{t+\ell}) \sim \ell^{-g}$ ,  $0 < g < 1$ : metaorders are split [Bouchaud, Gefen, Potters & Wyart \(2004\)](#)
  - ▶ Yet prices are close to a random walk: the propagator model  $p_t = \sum_{s < t} G(t-s) \varepsilon_s + \text{noise}$  with a decaying kernel  $G$  reconciles the two
- Arrival times cluster: a Hawkes process has intensity  $\lambda(t) = \mu + \int_{-\infty}^t \phi(t-s) dN_s$  [Bacry, Mastromatteo & Muzy \(2015\)](#)
  - ▶ Estimated by maximum likelihood; the branching ratio  $\|\phi\|_1 = \int \phi$  is the share of events triggered by earlier events (endogeneity), stationary if  $< 1$
- Links: durations and clustering of Chapter 8, intraday volatility of Chapter 9, the concave impact below

## Metaorders and the Square-Root Law

- A **metaorder**: a large order split into many small trades over hours or days
- Empirical law for the peak impact (price move from the first to the last trade of the metaorder) [Tóth et al. \(2011\)](#), [Bouchaud, Bonart, Donier & Gould \(2018\)](#)
  - ▶  $\Delta P \approx Y \sigma_{\text{day}} \sqrt{Q/V_{\text{day}}}$ , with  $Q$  the metaorder size,  $V_{\text{day}}$  the daily volume,  $Y$  of order 1
- Concave: trading 1% of daily volume moves the price by about  $0.1 \sigma$ , not  $0.01 \sigma$ : liquidity near the price is scarce and refills slowly (Section 2)
- Cost: if impact builds as  $I(q) = Y \sigma \sqrt{q/V}$  during execution, the average price paid is  $\frac{1}{Q} \int_0^Q I(q) dq = \frac{2}{3} I(Q)$  (the shortfall, before spread and fees); in currency, times  $Q$  and the price
- Testing it needs the traders' own order records; public bars allow only an indirect look (next slide)

## Volume and Price Moves in SPY Bars



- 115 052 five-minute bars in 20 equal-size groups by relative volume (bar volume / median volume at that time of day)

## Reading the Volume–Move Relation

- ▣ Mean  $|r|$ , scaled by the day's volatility, rises with relative volume with a slope of 0.27 on log-log axes
- ▣ Concave, as the square-root law predicts, but flatter than  $1/2$
- ▣ Not a causal elasticity: **simultaneity**, news drives both volume and  $|r|$ ; and opposite trades in a bar net out
  - ▶ An instrument for volume must move volume but not  $|r|$  directly; seminar B6 tries the same bar's volume on the previous day and questions the exclusion restriction
- ▣ For execution costs use the metaorder law, calibrated on one's own trades

## Optimal Execution: the Trade-Off

- Sell  $X$  shares within  $T$  days, in  $N$  steps of length  $\tau = T/N$ ; holdings  $x_k$  with  $x_0 = X$ ,  $x_N = 0$ ; trades  $n_k = x_{k-1} - x_k$   
Almgren & Chriss (2001)
- Price dynamics:  $S_k = S_{k-1} + \sigma\sqrt{\tau}\xi_k - \gamma n_k$ ; execution price  $\tilde{S}_k = S_{k-1} - \eta n_k/\tau$ 
  - ▶  $\gamma$ : permanent impact;  $\eta$ : temporary impact, paid only by the trade itself
- **Trading fast**: high temporary impact, low price risk; **trading slowly**: the opposite
- Objective: minimise  $\mathbb{E}[C] + \lambda \text{Var}[C]$ , with  $C$  the implementation cost and  $\lambda$  the risk aversion

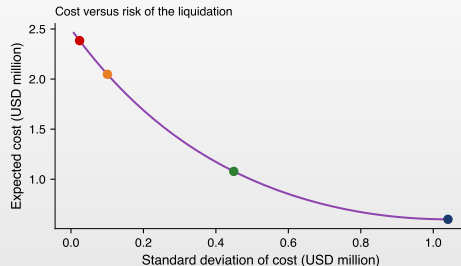
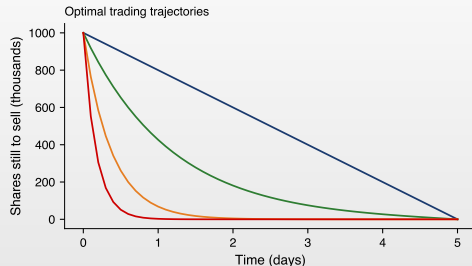
## Deriving the Almgren–Chriss Trajectory

- Continuous time, holdings  $x(t)$ , trading rate  $-\dot{x}$ : temporary impact cost  $\eta \int_0^T \dot{x}^2 dt$ , price-risk variance  $\sigma^2 \int_0^T x^2 dt$ 
  - Minimise  $\int_0^T (\eta \dot{x}^2 + \lambda \sigma^2 x^2) dt$  subject to  $x(0) = X$ ,  $x(T) = 0$  (the permanent impact  $\frac{1}{2}\gamma X^2$  does not depend on the path)
- Euler–Lagrange for  $F(x, \dot{x}) = \eta \dot{x}^2 + \lambda \sigma^2 x^2$ :  $\frac{d}{dt} \partial F / \partial \dot{x} = \partial F / \partial x \Rightarrow \ddot{x} = \kappa^2 x$ ,  $\kappa^2 = \lambda \sigma^2 / \eta$
- General solution  $Ae^{\kappa t} + Be^{-\kappa t}$ ; the boundary conditions give  $x(t) = X \sinh(\kappa(T - t)) / \sinh(\kappa T)$
- In discrete time  $\kappa^2$  becomes  $\frac{2}{\tau^2} (\cosh \kappa \tau - 1) = \lambda \sigma^2 / \tilde{\eta}$  (next slide); a resilient book with transient impact changes the shape [Obizhaeva & Wang \(2013\)](#)

## Almgren–Chriss: the Optimal Trajectory

- $x_k = X \frac{\sinh(\kappa(T - t_k))}{\sinh(\kappa T)}, \quad \frac{2}{\tau^2} (\cosh(\kappa\tau) - 1) = \tilde{\kappa}^2 = \frac{\lambda\sigma^2}{\tilde{\eta}}, \quad \tilde{\eta} = \eta - \frac{\gamma\tau}{2}$
- $\mathbb{E}[C] = \frac{1}{2}\gamma X^2 + \frac{\tilde{\eta}}{\tau} \sum_{k=1}^N n_k^2, \quad \text{Var}[C] = \sigma^2\tau \sum_{k=1}^N x_k^2$
- $\lambda = 0$ :  $\kappa = 0$  and  $x_k$  falls linearly (trade evenly, the time-weighted schedule)
  - Large  $\lambda$ : most of the order is traded at the start, to cut the exposure to price risk
- $1/\kappa$  is the “half-life” of the trade: the time scale over which the position is worked off

## Optimal Trajectories and the Efficient Frontier



— Risk-neutral,  $\lambda = 0$  (sell evenly)      —  $\lambda = 2e-5$       — Efficient frontier (N = 5 days)  
 —  $\lambda = 2e-6$       —  $\lambda = 1e-4$  (very risk averse)

- Numerical example of [Almgren & Chriss \(2001\)](#):  $X = 10^6$  shares,  $S_0 = 50$  USD,  $T = 5$  days,  $\sigma = 0.95$  USD per day<sup>1/2</sup>,  $\eta = 2.5 \cdot 10^{-6}$ ,  $\gamma = 2.5 \cdot 10^{-7}$

## Reading the Execution Chart

| Strategy (5 daily steps)    | Expected cost    | Standard deviation |
|-----------------------------|------------------|--------------------|
| Even selling, $\lambda = 0$ | 0.60 million USD | 1.04 million USD   |
| $\lambda = 2 \cdot 10^{-6}$ | 1.08             | 0.45               |
| $\lambda = 2 \cdot 10^{-5}$ | 2.05             | 0.10               |
| $\lambda = 10^{-4}$         | 2.38             | 0.02               |

- With  $\lambda = 2 \cdot 10^{-6}$  the first day sells 57% of the position; 428 599 shares remain
- Moving from even selling to  $\lambda = 2 \cdot 10^{-6}$  cuts the standard deviation from 1.04 to 0.45 million USD, for 0.48 million USD more expected cost
- The frontier is convex: the first units of risk reduction are cheap, the last ones expensive

## Execution Benchmarks

- ▣ **Implementation shortfall:** the difference between the paper portfolio at the decision price and the executed one [Perold \(1988\)](#)
- ▣ VWAP (Volume-Weighted Average Price): trade in proportion to the expected volume curve (the U-shape of Section 5)
- ▣ TWAP (Time-Weighted Average Price): equal amounts in each interval, the  $\lambda = 0$  solution
- ▣ A participation algorithm (a fixed share of market volume) ignores price
  - ▶ This is the algorithm at the heart of the Flash Crash (next section)

## Recap: Information, Impact and Execution

- ▣ Adverse selection alone creates a spread; Kyle's  $\lambda = \sigma_v / (2\sigma_u)$  is a regression slope, estimable from order flow
- ▣ Impact is concave in size: the square-root law for metaorders
- ▣ Optimal execution trades off expected impact cost against price risk
- ▣ Risk-averse traders front-load; risk-neutral traders spread the order evenly

## High-Frequency Trading: the New Market Makers



A microwave relay tower (Mojave, California); similar line-of-sight links carry trading signals between Chicago and New Jersey

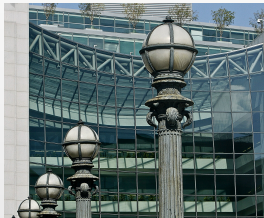
Photo: Tony Webster (2016); CC BY-SA 2.0; [Wikimedia Commons](#)

- HFT: algorithms that trade in micro- to milliseconds and end the day flat; speed from co-location and microwave links (faster through air than light in fibre)
- Low latency speeds up price discovery and narrows spreads [Hasbrouck & Saar \(2013\)](#)
- One HFT firm made markets on Chi-X and Euronext, mostly passively: it earned the spread and lost on positions [Menkveld \(2013\)](#)
  - ▶ Order flow and latency become “information” [O’Hara \(2015\)](#); market makers as intermediaries: [Pele \(2025\)](#)
- HFT liquidity comes and leaves fast: plentiful in calm markets, fragile under stress

## Optimal Market Making: Avellaneda–Stoikov

- A dealer with inventory  $q$  maximises expected exponential (CARA) utility of terminal wealth, risk aversion  $\gamma$ ; mid-price  $s_t$  is a Brownian motion with volatility  $\sigma$  [Avellaneda & Stoikov \(2008\)](#)
  - ▶ Market orders hit a quote at distance  $\delta$  from the mid with intensity  $Ae^{-k\delta}$
- Reservation (indifference) price:  $r = s - q\gamma\sigma^2(T - t)$ : a long dealer ( $q > 0$ ) values the asset below the mid
- Approximate optimal total spread around  $r$  (from an expansion of the value function, not the exact solution):  
 $\delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma} \ln\left(1 + \frac{\gamma}{k}\right)$
- Inventory skews both quotes: this formalises the inventory component of the spread (Section 3); the  $\ln(1 + \gamma/k)$  term prices the thinness of the order flow

## The Flash Crash of 6 May 2010

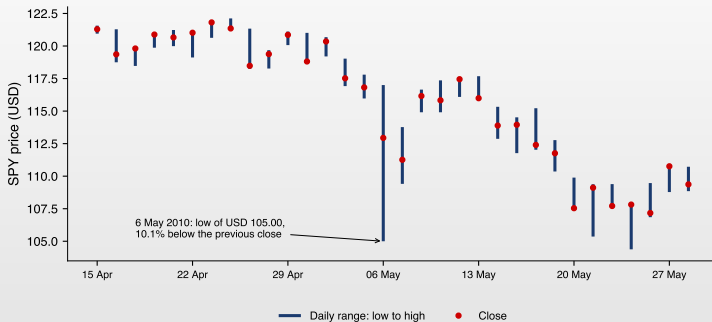


Headquarters of the US Securities and Exchange  
Commission, Washington

Photo: David (dbking) (2008); CC BY 2.0; Wikimedia  
[Commons](#)

- Report of the US futures (CFTC) and securities (SEC) regulators [CFTC & SEC \(2010\)](#)
- At 2:32 p.m. a mutual fund complex started selling 75 000 E-Mini S&P 500 futures (about USD 4.1 billion)
  - ▶ The algorithm targeted 9% of the previous minute's volume, "without regard to price or time"
- By 2:45:28 p.m. buy-side depth in the E-Mini had fallen below 1% of its morning level
- Over 20 000 trades in more than 300 securities were executed more than 60% away from their earlier prices, some at a penny

## SPY Around the Flash Crash

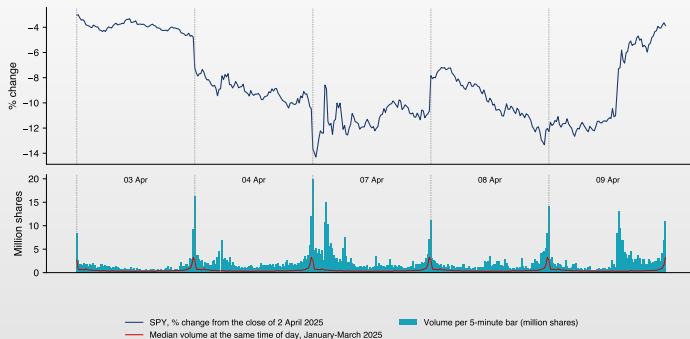


□ Daily low-high range and close, 15 April – 28 May 2010 (prices not adjusted)

## Reading the Flash Crash

- ▣ SPY low of USD 105.00: 10.1% below the previous close of 116.82; it still closed only 3.3% lower
- ▣ Daily volume was 3.0 times the median of the previous three weeks
- ▣ HFT firms did not trigger the crash, but their reaction to the selling pressure amplified it [Kirilenko, Kyle, Samadi & Tuzun \(2017\)](#)
  - ▶ A 5-second pause in E-Mini trading at 2:45:28 p.m. let buyers return [CFTC & SEC \(2010\)](#)
- ▣ Responses: single-stock circuit breakers (a 5-minute pause after a 10% move in 5 minutes) and clearer rules for cancelling erroneous trades [CFTC & SEC \(2010\)](#)

## A Recent Stress Episode: April 2025



- SPY, 5-minute bars, 3–9 April 2025; bottom panel: bar volume against the median volume at the same time of day in January–March 2025

## Reading April 2025

- 7 April: intraday range of 8.6%; one 5-minute bar rose 4.28%, and a bar ten minutes later fell 2.92%
  - ▶ Realised volatility of the day: 6.7%, against a sample median of 0.61%
- 9 April: +10.0% from the open to the close
- Over the five days, volume was 4.6 times its usual level at the same times of day
- The bars document sharp moves, heavy trading and a high price impact per dollar (Section 5); they cannot establish price continuity, the absence of trading halts or order-book depth

## The Speed Race and Frequent Batch Auctions

- In a continuous order book, correlations between related markets break down at millisecond horizons [Budish, Cramton & Shim \(2015\)](#)
  - ▶ This creates arbitrage opportunities that go to the fastest trader: an arms race in speed
- The cost of the race is borne by investors through wider spreads and thinner books
- Proposal: **frequent batch auctions**, e.g. every second, at a single price
- Competition moves from speed to price: narrower spreads and deeper markets in the model

## Crypto Venues and Arbitrage

- 24/7 trading with no auctions; the same asset trades on many CEX order books at once, and on DEX pools priced by a formula (AMM, Chapter 16)
  - ▶ The same Bitcoin can trade at different prices on different exchanges [Makarov & Schoar \(2020\)](#)
- Price gaps are much larger **across countries** than within a country
  - ▶ Capital controls limit the movement of arbitrage capital
- A common component of signed volume across exchanges explains 80% of Bitcoin returns [Makarov & Schoar \(2020\)](#)
- Liquidity: high-low estimators track its time variation, Amihud-type ratios its level [Brauneis, Mestel, Riordan & Theissen \(2021\)](#); it can be forecast from market data [Agakishiev, Härdle & Pele \(2025\)](#)

## Price Discovery across Venues: Information Shares

- Two prices of one asset,  $y_t = (p_{1t}, p_{2t})'$ , are  $I(1)$  and cointegrated with  $\beta = (1, -1)'$ ; VECM (vector error-correction model):
  - ▶  $\Delta y_t = \alpha \beta' y_{t-1} + \sum_{i=1}^p \Gamma_i \Delta y_{t-i} + e_t$ ,  $\text{Var}(e_t) = \Omega$  [Johansen \(1991\)](#)
  - ▶ Common efficient price: the stochastic trend with loadings  $\psi' \propto \alpha'_\perp$ , where  $\alpha'_\perp \alpha = 0$
- **Hasbrouck information share**: venue  $j$ 's share of the variance of the common-trend innovation,  $([\psi' F]_j)^2 / \psi' \Omega \psi$ ,  $F$  the Cholesky factor of  $\Omega$  [Hasbrouck \(1995\)](#)
  - ▶ Depends on the ordering when  $\Omega$  is not diagonal: report the bounds over both orderings
- **Gonzalo–Granger component share**:  $\alpha_\perp$  normalised to sum 1; in  $[0, 1]$  only if the two adjustment coefficients have opposite signs [Gonzalo & Granger \(1995\)](#)
- A price that does **not** adjust ( $\alpha_j = 0$ , weak exogeneity) leads price discovery; the inference is a Wald test of  $\alpha_j = 0$ , and non-rejection does not prove  $\alpha_j = 0$

## The Bucharest Stock Exchange



Offices of the Bucharest Stock Exchange

Photo: [Za.kurek \(2014\)](#); CC BY-SA 4.0; [Wikimedia](#)

[Commons](#)

- An electronic limit order book, as in the rest of this chapter [BVB \(history\)](#)
- Emerging-market status (FTSE Russell) since September 2020; Banca Transilvania and Nuclearelectrica were the first stocks in FTSE emerging-market indices
- A thin market: herding and bubbles in illiquid markets were studied on the BVB by [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
  - ▶ Bubbles and crashes are the subject of Chapter 17

## BVB Liquidity: What the Data Say

- ▣ Amihud ratio of Banca Transilvania: 24.7 bp per USD 1 million; of Transelectrica: 1024.0; of SPY: 0.0016
- ▣ The BET ETF trades less than Banca Transilvania: 212.9 bp per USD 1 million
- ▣ Illiquidity has fallen since 2017, but the BVB remains thousands of times less liquid than US large caps
  - ▶ Daily spread estimates hide this: they rank BVB stocks with US stocks, because volatilities are similar
- ▣ Monthly market returns fall when illiquidity rises unexpectedly, on the BVB as in the US and in crypto (t-statistics -4.0, -3.8, -4.1; seminar Part C)

## Price Discovery on the BVB: the BET ETF and the BET-TR Index

- ▣ Daily log closes of the Patria-TV BET ETF (it reinvests dividends) and the BET-TR index, 495 common trading days, September 2024 – September 2026; the basis has s.d. 0.84%; BET-TR is computed from its constituents' prices, not traded on a venue
- ▣ Johansen trace test: 49.6 for  $r = 0$  (5% critical value 15.49), 0.10 for  $r \leq 1$  (3.84): one cointegrating vector; VECM with 1 lag (BIC)
- ▣ Adjustment: ETF  $\alpha = -0.12$  ( $t = -2.11$ ); index  $\alpha = -0.03$  ( $t = -0.63$ )
  - ▶ The index adjustment is not distinguishable from zero, the ETF's is: suggestive evidence that the ETF adjusts toward the index, with wide bootstrap intervals (seminar B8)
- ▣ Information share of the index: between 0.25 and 0.98; residual correlation 0.93. Component share of the index 1.39, outside  $[0, 1]$  because both  $\alpha$  are negative
- ▣ Lesson: with daily closes the innovations are almost collinear, so the bounds are wide; price discovery needs intraday data (seminar B8)

## Recap: Stress, Crypto and the BVB

- ▣ HFT liquidity is cheap in normal times and fragile under stress (2010); price impact per dollar also rose in April 2025; market design can reduce the value of pure speed
- ▣ Cross-venue price discovery is a cointegration problem; daily data give wide information-share bounds
- ▣ Concentrated trading: two stocks carry more than half of the blue-chip traded value
- ▣ Use traded-value measures (Amihud), not daily spread proxies, to compare the BVB with other markets
- ▣ For portfolio decisions, the illiquidity of the BVB is a cost and a risk (Chapter 4 and seminar Part C)

## An Open Question: Do Price-Based Proxies Measure Trading Costs?

- Where trades and quotes are unavailable (BVB history, many crypto venues), costs are estimated from daily OHLC bars
  - ▶ Jahan-Parvar & Zikes (2023): low-frequency estimators track effective spreads only when the spread is large relative to volatility; otherwise volatility dominates
  - ▶ Ardia, Guidotti & Kroencke (2024): EDGE, an estimator from open, high, low and close prices, unbiased and efficient under their assumptions
- Formal question:  $\hat{S}_i = S_i + b(\sigma_i) + \varepsilon_i$  across assets  $i$ 
  - ▶  $H_0$ :  $b \equiv 0$  (the proxy ranks assets by cost) against  $H_1$ :  $b$  increasing in  $\sigma_i$
  - ▶ Identification needs a cost benchmark measured independently of the proxy: effective spreads from matched trades and quotes; Amihud measures price impact, not the spread, and itself contains  $|r|$
- Open: the spread-to-volatility ratio is lowest exactly where proxies are needed most (volatile, thinly traded markets)

## The Discovery Loop, Step by Step (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: collect validation studies, then check every DOI on [Crossref](#) (the title must match)
  - ▶ Find studies that validate Roll, Corwin-Schultz, Abdi-Ranaldo and EDGE against effective spreads from trades and quotes; report the benchmark, the correlation in levels and in changes, the markets, and the DOI.
- 2. **Competing hypotheses**, each with a cross-sectional prediction
  - ▶ Give rival explanations for why Corwin-Schultz ranks BVB blue chips next to US large caps, and for each the predicted Spearman correlation of the estimate with volatility, with the Amihud ratio and with traded value.
- 3. **Data and code**: replicate the chapter's numbers first
  - ▶ Write Python that reads `data/market/<symbol>.csv` for the 25 assets of this chapter and computes Corwin-Schultz exactly as in the paper, with the overnight adjustment and negative two-day estimates set to zero. Target medians: US 53, BVB 54, crypto 124 bp

## The Discovery Loop, Step by Step (2/2)

- 4. **Robustness**, pre-registered
  - ▶ Write a pre-registered plan: Spearman and partial Spearman correlations, bootstrap over assets, leave-one-market-out (US, BVB, crypto), a second window 2022-2024, and EDGE and Abdi-Ranaldo in place of Corwin-Schultz.
- 5. **Critique**: ask the assistant to attack the benchmark itself
  - ▶ Act as a referee: argue that the Amihud ratio is not a cost measure either (it scales with volume and with volatility); which benchmark would identify the spread, and what would it cost to obtain?
- 6. **Report**: every estimator and window, bootstrap intervals, the benchmark's limits, code and the AI log

## What the Researcher Must Check

- **Tick-size benchmark:** no **quoted** spread is below one tick (effective spreads can be); compare every estimate with  $0.01/P$  (0.21 bp for SPY) as a plausibility check, not a validation
- **Volatility contamination:** test: partial correlation with the benchmark controlling for volatility, and the 5-minute versus daily comparison of Section 3
- **Truncation:** negative estimates set to zero bias the mean upwards; report the share truncated and a median
- **Units and venues:** lei converted at the BNR rate; crypto volume summed over exchanges, some with inflated volume; test: rerun on a subset of venues
- **Small, clustered cross-section:** 25 assets in 3 markets are not exchangeable, and 3 clusters are too few for cluster-robust inference; report the asset bootstrap as conditional on independence, plus a within-market bootstrap and leave-one-market-out
- **Mistyped formulas:** check the constants of the estimator against the paper and run a unit test on simulated prices with a known spread

## Mini-Case: Cost or Volatility?

- Replication: median Corwin–Schultz estimates US 53, BVB 54, crypto 124 bp (Section 3)
- A typical assistant proposal: “the estimates are similar, so BVB blue chips are about as liquid as US large caps”.  
Diagnostic: compare the proxy’s rank correlation with volatility and with the Amihud price-impact ratio
- Test on the 25 assets, September 2024 – September 2026 (Spearman rank correlations)
  - ▶ With daily volatility:  $\rho = 0.94$  ( $p < 0.001$ ); with the Amihud ratio:  $\rho = -0.12$  ( $p = 0.58$ )
  - ▶ Difference 1.06: 95% CI [0.58, 1.44] resampling assets as if independent, [0.71, 1.43] resampling within markets; partial  $\rho$  with Amihud given volatility: 0.27
- Interpretation: daily Corwin–Schultz ranks track volatility, not price impact, consistent with the contamination  
[Jahan-Parvar & Zikes \(2023\)](#) predict for a low spread-to-volatility ratio; a diagnostic, not a test of  $H_0$ , which needs matched effective spreads
- Code: `Quantlets/Ch_10/ai_discovery_case.py`

## Your Turn: A Project Seed

- Question: which estimator from daily bars, if any, measures the level of trading costs on the BVB and for crypto-assets?
- Replicate first: SPY Corwin–Schultz from daily bars 26.5 bp versus 3.11 bp from 5-minute bars and a tick floor of 0.21 bp (Section 3)
- Deliverables
  - ▶ A pre-registration (hypotheses, estimators, benchmark, tests, corrections) dated before the first run
  - ▶ EDGE implemented exactly as in [Ardia, Guidotti & Kroencke \(2024\)](#), validated on simulated prices with a known spread, then compared with the other estimators on the 25 assets
  - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in AI\_USE.md and every error the assistant made in AI\_ERRORS.md:  
[github.com/danpele/MFM/tree/main/project](https://github.com/danpele/MFM/tree/main/project)

## Summary

- ▣ Prices are formed in limit order books: resting limit orders supply liquidity, market orders and marketable limit orders consume it
- ▣ The spread pays for processing, inventory and adverse selection; Glosten–Milgrom and Kyle model the last one
- ▣ Daily spread proxies are contaminated by volatility when the spread is small relative to volatility, as for the liquid assets here; the Amihud ratio separates markets by orders of magnitude
- ▣ Liquidity follows the clock (U-shape) and the VIX; it dries up under stress (2010) and price impact rises (April 2025)
- ▣ Impact is concave in size; optimal execution balances impact against price risk
- ▣ Inference matters: the Roll covariance is biased in small samples, structural models (MRR, Hasbrouck) need signed trades, price discovery is a VECM problem

## Which Liquidity Measure When?

| Available data                        | Suggested measure                                                                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trades and quotes with timestamps     | Effective and realised spread; MRR by GMM; Hasbrouck VAR; PIN                                                                                                   |
| Prices of one asset on several venues | VECM; information-share bounds and component shares                                                                                                             |
| Intraday bars (1–5 minutes)           | Corwin–Schultz or Abdi–Rinaldo per day; intraday Amihud                                                                                                         |
| Daily prices and volumes              | Amihud ratio (price impact) across assets; EDGE for spread levels only if validated against effective spreads; high-low estimators mainly for changes over time |
| Own executions                        | Implementation shortfall; calibrate the square-root law                                                                                                         |
| Theory, what-if analysis              | Simulated order books; Glosten–Milgrom, Kyle, Almgren–Chriss                                                                                                    |

## Coming Next: Continuous-Time Models (Chapter 11)

- ▣ From trades and order books back to smooth models of prices: Brownian motion and Itô calculus
- ▣ At high frequency, microstructure noise contaminates volatility estimates: realised volatility (Chapter 9) needs care
- ▣ Further reading: Hasbrouck (2007), Bouchaud, Bonart, Donier & Gould (2018), Harris (2003), Franke, Härdle & Hafner (2019)

## Further Study on Quantinar

### Advanced courses on quantinar.com

 Applied Computational Finance

### Code and data

 Quantlets: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_10](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_10)

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